SUMMATIVE ASSESSMENT – 1 (SA-1) MODEL PAPER – SET 1

CLASS X · MATHEMATICS · AP State Syllabus 2026-27 · SSC Blueprint 15E & 16E Pattern

BLUEPRINT – SA-1 MODEL PAPER – SET 1

SUBJECT: MATHEMATICS	CLASS: X	TIME: 3 Hrs 15 Min	MAX. MARKS: 100

Syllabus: Chapters 1 – 11 of Class X Mathematics (AP SCERT, 2026-27) – Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations, Arithmetic Progressions, Triangles, Coordinate Geometry, Statistics, Introduction to Trigonometry, Applications of Trigonometry, Probability. Pattern: SSC Public Examinations 2026-27 Blueprint & Model Paper (15E & 16E).

1. Weightage to Academic Standards (Ability)

ABILITY	AWARENESS (60%)	SENSITIVITY (20%)	CREATIVITY (20%)	TOTAL
MARKS	60	20	20	100

2. Weightage to Form of Questions

FORM OF QUESTION	Essay / Long Ans.	Short Answer	Very Short Ans.	Objective	TOTAL
No. of questions	5	8	8	12	33
Marks allotted	8	4	2	1	100
Estimated time	70 min	60 min	30 min	20 min	180 + 15 min

3. Ability-wise Distribution of Questions

ABILITY	1 Mark	2 Marks	4 Marks	8 Marks	Total Marks
Awareness	4	4	6	3	60
Sensitivity	4	2	1	1	20
Creativity	4	2	1	1	20
Total Questions	12	8	8	5	100

4. Chapter-wise Weightage

S.No	UNIT / CHAPTER	MARKS
1	Real Numbers	1+8 = 9
2	Polynomials	1+2+4 = 7
3	Pair of Linear Equations	1+2+8 = 11
4	Quadratic Equations	1+2+4+(8) = 7 + (8)
5	Arithmetic Progressions	1+4+(8) = 5 + (8)
6	Triangles	1+1+2+4+(8) = 8 + (8)
7	Introduction to Trigonometry	1+2+4+(8) = 7 + (8)
8	Applications of Trigonometry	1+2+(8) = 3 + (8)
9	Coordinate Geometry	1+2+4+8 = 15
10	Statistics	1+2+4+8 = 15
11	Probability	1+4+8 = 13
	TOTAL	**100 + (40)**

Note: Marks within ( ) are the internal-choice (OR) questions of Section IV. The marks of the chapters that are not part of SA-1 in the SSC blueprint (Circles, Areas Related to Circles, Surface Areas & Volumes) are redistributed among Chapters 1 – 11; the ability-wise (60 : 20 : 20) and form-wise weightage are exactly as in the SSC blueprint.

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5. Question-wise Chapter Distribution and Domain (Questions 1 to 33)

Q.No	Chapter / Topic	Domain	Marks
1	Real Numbers	Awareness	1
2	Polynomials	Creativity	1
3	Pair of Linear Equations	Creativity	1
4	Quadratic Equations	Creativity	1
5	Arithmetic Progressions	Sensitivity	1
6	Triangles	Awareness	1
7	Introduction to Trigonometry	Sensitivity	1
8	Applications of Trigonometry	Sensitivity	1
9	Coordinate Geometry	Creativity	1
10	Statistics	Awareness	1
11	Probability	Awareness	1
12	Triangles	Sensitivity	1
13	Polynomials	Creativity	2
14	Quadratic Equations	Sensitivity	2
15	Triangles	Awareness	2
16	Coordinate Geometry	Sensitivity	2
17	Introduction to Trigonometry	Awareness	2
18	Statistics	Awareness	2
19	Pair of Linear Equations	Awareness	2
20	Applications of Trigonometry	Creativity	2
21	Polynomials	Awareness	4
22	Quadratic Equations	Awareness	4
23	Arithmetic Progressions	Awareness	4
24	Introduction to Trigonometry	Awareness	4
25	Triangles	Sensitivity	4
26	Coordinate Geometry	Awareness	4
27	Statistics	Awareness	4
28	Probability	Creativity	4
29a / 29b	(a) Real Numbers (b) Arithmetic Progressions	Creativity	8
30a / 30b	(a) Statistics (b) Triangles	Awareness	8
31a / 31b	(a) Coordinate Geometry (b) Introduction to Trigonometry	Sensitivity	8
32a / 32b	(a) Probability (b) Quadratic Equations	Awareness	8
33a / 33b	(a) Pair of Linear Equations (b) Applications of Trigonometry	Awareness	8

ABILITIES AND THEIR OBJECTIVES AS PER PARAKH NORMS

I) AWARENESS: Knowledge / understanding of the construct / understanding of the process (resembles old objectives Knowledge, Understanding and Application).

II) SENSITIVITY: Conflict resolution / open-mindedness / collaboration (resembles old objective Analysis).

III) CREATIVITY: Generate innovative ideas / flexibility / fluency / exploration / combining ideas and concepts (resembles old objectives Evaluation and Creativity).

NOTE: Overall, the chapter-wise and ability-wise weightage is followed. There is no rigid rule that a question with a particular number must come from a particular chapter.

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SUMMATIVE ASSESSMENT – 1 : MODEL QUESTION PAPER – SET 1

MATHEMATICS – CLASS X (English Medium, AP State Syllabus 2026-27)

Time : 3 Hrs 15 Min (15 min reading + 3 hrs writing) **Max. Marks : 100**

General Instructions: *(i) Answer all the questions. (ii) Write the answers neatly in the answer booklet. (iii) The paper has four Sections (I, II, III, IV). (iv) Use of calculators is not permitted. (v) Draw neat, labelled figures wherever necessary.*

SECTION – I 12 × 1 = 12 Marks

Note : *i) Answer all the questions. ii) Each question carries 1 mark. iii) Choose the correct option and write its letter (A/B/C/D) in the answer booklet.*

1. Which of the following rational numbers has a terminating decimal expansion? *[Awareness · Real Numbers]*

(A) 7/30 (B) 11/40 (C) 5/12 (D) 9/14

2. If the sum of the zeroes of the polynomial $kx^2 + 2x + 3k$ is equal to their product, then k = *[Creativity · Polynomials]*

(A) 1/3 (B) −1/3 (C) 2/3 (D) −2/3

3. The value of k for which the pair of equations $2x + 3y = 7$ and $(k - 1)x + (k + 2)y = 3k$ has infinitely many solutions is *[Creativity · Pair of Linear Equations]*

(A) 5 (B) 6 (C) 7 (D) 8

4. The values of k for which the equation $x^2 + kx + 16 = 0$ has equal roots are *[Creativity · Quadratic Equations]*

(A) 4 (B) ±4 (C) ±8 (D) 8

5. Ravi saves Rs. 10 on the first day and increases his saving by Rs. 5 every next day. The amount he saves on the 8th day is *[Sensitivity · Arithmetic Progressions]*

(A) Rs. 40 (B) Rs. 45 (C) Rs. 50 (D) Rs. 55

6. In ΔABC, DE ∥ BC where D is on AB and E is on AC. If AD = 2 cm and DB = 3 cm, then AE : EC = *[Awareness · Triangles]*

(A) 2 : 5 (B) 3 : 2 (C) 2 : 3 (D) 3 : 5

7. If $5 \tan\theta = 4$, then $(5 \sin\theta - 3 \cos\theta)/(5 \sin\theta + 2 \cos\theta)$ = *[Sensitivity · Introduction to Trigonometry]*

(A) 1/6 (B) 1/5 (C) 1/7 (D) 1/3

8. A ladder 10 m long reaches a window $5\sqrt{3}$ m above the ground. The angle made by the ladder with the ground is *[Sensitivity · Applications of Trigonometry]*

(A) 30° (B) 45° (C) 60° (D) 90°

9. If (a, b) is the mid-point of the line segment joining (10, −6) and (k, 4) and $a - 2b = 18$, then k = *[Creativity · Coordinate Geometry]*

(A) 22 (B) 20 (C) 18 (D) 26

10. Which of the following is NOT a measure of central tendency? *[Awareness · Statistics]*

(A) Mean (B) Median (C) Mode (D) Range

11. A die is thrown once. The probability of getting a prime number is *[Awareness · Probability]*

(A) 1/6 (B) 1/3 (C) 1/2 (D) 2/3

12. In ΔABC, DE ∥ BC with AD = x, DB = x − 2, AE = x + 2 and EC = x − 1. Then x = *[Sensitivity · Triangles]*

(A) 3 (B) 4 (C) 5 (D) 6

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SECTION – II 8 × 2 = 16 Marks

Note : *i) Answer all the questions. ii) Each question carries 2 marks.*

13. Find a quadratic polynomial whose sum of zeroes is −3 and product of zeroes is 2. Also find its zeroes. *[Creativity · Polynomials]*

14. Sita says that the equation $x^2 + 3x + 5 = 0$ has real roots, but Gita disagrees. Who is correct? Give reason. *[Sensitivity · Quadratic Equations]*

15. State the SSS criterion for similarity of two triangles. *[Awareness · Triangles]*

16. Are the points A(1, 5), B(2, 3) and C(−2, −11) collinear? Justify your answer using the distance formula. *[Sensitivity · Coordinate Geometry]*

17. Evaluate: $\sin 60° \cos 30° + \sin 30° \cos 60°$. *[Awareness · Introduction to Trigonometry]*

18. Write the formula for the mode of grouped data and explain the meaning of each symbol used. *[Awareness · Statistics]*

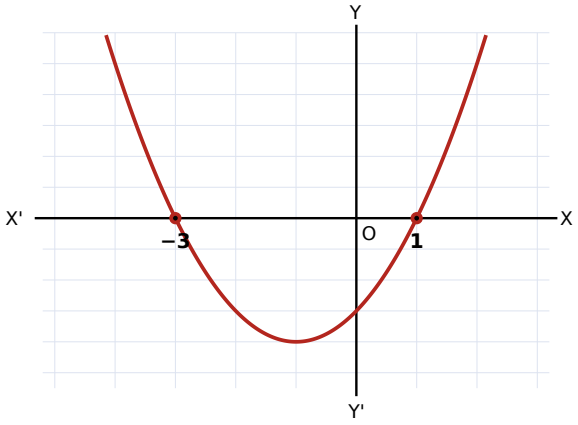
19. Solve by the elimination method: x + y = 14 and x − y = 4. *[Awareness · Pair of Linear Equations]*

20. A kite is flying at a height of 60 m above the ground. The string attached to the kite is tied to a point on the ground and makes an angle of 60° with the ground. Draw a rough figure and find the length of the string (assume no slack). *[Creativity · Applications of Trigonometry]*

SECTION – III 8 × 4 = 32 Marks

Note : *i) Answer all the questions. ii) Each question carries 4 marks.*

21. Observe the graph of a polynomial given below and answer the questions. *[Awareness · Polynomials]*



i) How many zeroes does the polynomial have?
ii) Write the zeroes of the polynomial.
iii) What is the sum of the zeroes?
iv) Write the quadratic polynomial represented by the graph (leading coefficient 1).

22. The product of two consecutive positive integers is 306. Find the integers. *[Awareness · Quadratic Equations]*

23. Find the sum of all two-digit numbers which are divisible by 7. *[Awareness · Arithmetic Progressions]*

24. Prove that: $\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = 2 \operatorname{cosec}\theta$. *[Awareness · Introduction to Trigonometry]*

25. In ΔABC and ΔPQR, ∠A = ∠P = 50° and ∠B = ∠Q = 70°. Rahul says that the triangles are similar, but Salma says that the lengths of the sides are also needed. Resolve their conflict with a logical reason. If AB = 6 cm, BC = 8 cm and PQ = 9 cm, find QR. *[Sensitivity · Triangles]*

26. Find the coordinates of the points of trisection of the line segment joining A(2, −2) and B(−7, 4). *[Awareness · Coordinate Geometry]*

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27. The marks obtained by 40 students in a test are given below. Find the mean marks by the direct method. *[Awareness · Statistics]*

Marks	0–10	10–20	20–30	30–40	40–50
No. of students	5	8	12	10	5

28. A bag contains 5 red, 4 blue and 3 green balls. Write any four different probability questions based on drawing one ball at random from the bag and find the answer to each. *[Creativity · Probability]*

SECTION – IV 5 × 8 = 40 Marks

Note : *i) Answer all the questions. ii) Each question carries 8 marks. iii) There is an internal choice for each question.*

29.

a) Prove that $\sqrt{3}$ is an irrational number. Using it, justify that $5 + 2\sqrt{3}$ is also an irrational number. *[Creativity · Real Numbers]*

OR

b) The sum of the first n terms of an AP is $S_n = 3n^2 + 4n$. Find the first three terms of the AP, its nth term and the 20th term. *[Creativity · Arithmetic Progressions]*

30.

a) Find the median of the following distribution of marks of 50 students. *[Awareness · Statistics]*

Marks	0–10	10–20	20–30	30–40	40–50	50–60
No. of students	3	7	12	15	8	5

OR

b) Prove that if a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side (converse of the Basic Proportionality theorem). *[Awareness · Triangles]*

31.

a) The points A(1, 7), B(4, 2), C(−1, −1) and D(−4, 4) are joined in order. Rahul says ABCD is a rhombus and Sarita says it is a square. Resolve their conflict by showing, with calculations, which of them is correct. *[Sensitivity · Coordinate Geometry]*

OR

b) If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, show that $m^2 - n^2 = 4\sqrt{(mn)}$. *[Sensitivity · Introduction to Trigonometry]*

32.

a) A box contains 30 discs numbered 1 to 30. One disc is drawn at random. Find the probability that the number on it is: *[Awareness · Probability]*

i) a prime number
ii) a perfect square
iii) a multiple of 5
iv) neither prime nor composite

OR

b) A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train. *[Awareness · Quadratic Equations]*

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33.

a) Draw the graph of the pair of linear equations 2x + y = 6 and 2x − y = 2 and find the solution from the graph. Also find the area of the triangle formed by the two lines and the x-axis. *[Awareness · Pair of Linear Equations]*

OR

b) From the top of a 20 m high building, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. Find the height of the tower. *[Awareness · Applications of Trigonometry]*

— ALL THE BEST —

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